SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SAVEETHA UNIVERSTY

Department of Computer Science and Engineering

ASSIGNMENT QUESTIONS CO - 1

CSA04 -OPERATING SYSTEMS

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 30%

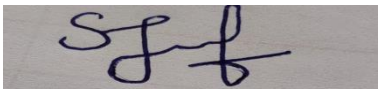
Dr.J.Velmurugan / Professor
Dept. of CSE

QUESTIONS: 5 **Total Marks:100**

Annexure A

Code of Conduct:

I S Jenefar (192511345) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Criteria	**Weightage (Marks)**	**Level 4 – Exemplary (Full Marks)**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 –** Beginning	**Total**
Technical Content Accuracy	**30**					
Problem Identification	**25**					
Use of Tools	**20**					
Communication	**15**					
Professionalism	**10**					
Total						

1. **Consider the set of 5 processes whose arrival time and burst time are given below**

Process Id	Arrival time	Burst time
P1	3	4
P2	5	3
P3	0	2
P4	5	1
P5	4	3

If the CPU scheduling policy is FCFS, calculate the average waiting time and average turn around time.

Answer:

Step 1: FCFS (First Come First Served) is non-preemptive. Processes are served strictly in order of arrival time. Arrival order: P3 (0), P1 (3), P5 (4), then P2 (5) and P4 (5). P2 and P4 arrive together, so the tie is broken by process number: P2 first, then P4.

Step 2: P3 runs 0-2. The CPU is idle from 2 to 3 because P1 has not arrived yet.

Step 3: P1 runs 3-7, P5 runs 7-10, P2 runs 10-13, P4 runs 13-14.

Gantt Chart:

P3	**Idle**	**P1**	**P5**	**P2**	**P4**

Time: 0 | 2 | 3 | 7 | 10 | 13 | 14

Calculation Table:

Process	**AT**	**BT**	**CT**	**TAT**	**WT**
P1	3	4	7	4	0
P2	5	3	13	8	5
P3	0	2	2	2	0
P4	5	1	14	9	8
P5	4	3	10	6	3

Turn Around Time (TAT) = Completion Time (CT) - Arrival Time (AT)

Waiting Time (WT) = Turn Around Time (TAT) - Burst Time (BT)

Total TAT = 4 + 8 + 2 + 9 + 6 = 29

Total WT = 0 + 5 + 0 + 8 + 3 = 16

Average Waiting Time = 16 / 5 = 3.2 units

Average Turn Around Time = 29 / 5 = 5.8 units

2. **Consider the set of 3 processes whose arrival time and burst time are given**

Process Id	Arrival time	Burst time
P1	0	2
P2	3	1
P3	5	6

If the CPU scheduling policy is FCFS, calculate the average waiting time and average turn around time.

Answer:

Step 1: FCFS serves processes in order of arrival: P1 (0), P2 (3), P3 (5).

Step 2: P1 runs 0-2. The CPU is idle from 2 to 3 (P2 has not arrived).

Step 3: P2 runs 3-4. The CPU is idle again from 4 to 5 (P3 has not arrived).

Step 4: P3 runs 5-11.

Gantt Chart:

P1	**Idle**	**P2**	**Idle**	**P3**

Time: 0 | 2 | 3 | 4 | 5 | 11

Calculation Table:

Process	**AT**	**BT**	**CT**	**TAT**	**WT**
P1	0	2	2	2	0
P2	3	1	4	1	0
P3	5	6	11	6	0

Turn Around Time (TAT) = Completion Time (CT) - Arrival Time (AT)

Waiting Time (WT) = Turn Around Time (TAT) - Burst Time (BT)

Total TAT = 2 + 1 + 6 = 9

Total WT = 0 + 0 + 0 = 0

Average Waiting Time = 0 / 3 = 0 units

Average Turn Around Time = 9 / 3 = 3 units

3. **Consider the set of 5 processes whose arrival time and burst time are given below**

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

If the CPU scheduling policy is SJF non-preemptive, calculate the average waiting time and average turn around time.

Answer:

Step 1: SJF non-preemptive: when the CPU becomes free, pick the arrived process with the smallest burst time. A running process is never interrupted.

Step 2: At t = 0 only P4 has arrived, so P4 runs 0-6.

Step 3: At t = 6 all of P1 (1), P2 (4), P3 (2), P5 (3) have arrived. Shortest first: P1 runs 6-7.

Step 4: Remaining are P2 (4), P3 (2), P5 (3). P3 runs 7-9, then P5 runs 9-12, then P2 runs 12-16.

Gantt Chart:

P4	**P1**	**P3**	**P5**	**P2**

Time: 0 | 6 | 7 | 9 | 12 | 16

Calculation Table:

Process	**AT**	**BT**	**CT**	**TAT**	**WT**
P1	3	1	7	4	3
P2	1	4	16	15	11
P3	4	2	9	5	3
P4	0	6	6	6	0
P5	2	3	12	10	7

Turn Around Time (TAT) = Completion Time (CT) - Arrival Time (AT)

Waiting Time (WT) = Turn Around Time (TAT) - Burst Time (BT)

Total TAT = 4 + 15 + 5 + 6 + 10 = 40

Total WT = 3 + 11 + 3 + 0 + 7 = 24

Average Waiting Time = 24 / 5 = 4.8 units

Average Turn Around Time = 40 / 5 = 8 units

4. Consider the set of 5 processes whose arrival time and burst time are given below

Process Id	Arrival time	Burst time
P1	3	1
P2	1	4
P3	4	2
P4	0	6
P5	2	3

If the CPU scheduling policy is SJF preemptive, calculate the average waiting time and average turn around time.

Answer:

Step 1: SJF preemptive (SRTF): at every arrival, the process with the shortest remaining time gets the CPU.

Step 2: t = 0-2: only P4 (6) is present, so it runs. At t = 2 P4 has 4 left. P5 arrives with 3, which is shorter, so P5 preempts P4.

Step 3: t = 2-3: P5 runs. At t = 3 P5 has 2 left. P1 arrives with 1, which is shorter, so P1 preempts P5.

Step 4: t = 3-4: P1 runs and finishes at 4. At t = 4, P3 arrives (2). Remaining times: P5 = 2, P3 = 2, P4 = 4, P2 = 4. P5 and P3 tie at 2, so the earlier arrival (P5) goes first.

Step 5: P5 runs 4-6 (done). P3 runs 6-8 (done).

Step 6: P4 (4) and P2 (4) tie, so the earlier arrival P4 runs 8-12, then P2 runs 12-16.

Gantt Chart:

P4	**P5**	**P1**	**P5**	**P3**	**P4**	**P2**

Time: 0 | 2 | 3 | 4 | 6 | 8 | 12 | 16

Calculation Table:

Process	**AT**	**BT**	**CT**	**TAT**	**WT**
P1	3	1	4	1	0
P2	1	4	16	15	11
P3	4	2	8	4	2
P4	0	6	12	12	6
P5	2	3	6	4	1

Turn Around Time (TAT) = Completion Time (CT) - Arrival Time (AT)

Waiting Time (WT) = Turn Around Time (TAT) - Burst Time (BT)

Total TAT = 1 + 15 + 4 + 12 + 4 = 36

Total WT = 0 + 11 + 2 + 6 + 1 = 20

Average Waiting Time = 20 / 5 = 4 units

Average Turn Around Time = 36 / 5 = 7.2 unit

5.Consider the set of 5 processes whose arrival time and burst time are given below

Process Id	Arrival time	Burst time
P1	0	5
P2	1	3
P3	2	1
P4	3	2
P5	4	3

If the CPU scheduling policy is Round Robin with time quantum = 2 unit, calculate the average waiting time and average turn around time.

Answer:

Step 1: Round Robin with time quantum = 2. Each process runs at most 2 units, then goes to the back of the ready queue. If a process arrives at the same moment another one is preempted, the new arrival is queued first.

Step 2: t = 0-2: P1 runs (rem 3). P2 arrived at 1 and P3 at 2. Queue: P2, P3, P1.

Step 3: t = 2-4: P2 runs (rem 1). P4 arrived at 3, P5 at 4. Queue: P3, P1, P4, P5, P2.

Step 4: t = 4-5: P3 runs for 1 and finishes. Queue: P1, P4, P5, P2.

Step 5: t = 5-7: P1 runs (rem 1). Queue: P4, P5, P2, P1.

Step 6: t = 7-9: P4 runs and finishes. Queue: P5, P2, P1.

Step 7: t = 9-11: P5 runs (rem 1). Queue: P2, P1, P5.

Step 8: t = 11-12: P2 finishes. t = 12-13: P1 finishes. t = 13-14: P5 finishes.

Gantt Chart:

P1	P2	P3	P1	P4	P5	P2	P1	P5

Time: 0 | 2 | 4 | 5 | 7 | 9 | 11 | 12 | 13 | 14

Calculation Table:

Process	AT	BT	CT	TAT	WT
P1	0	5	13	13	8
P2	1	3	12	11	8
P3	2	1	5	3	2
P4	3	2	9	6	4
P5	4	3	14	10	7

Turn Around Time (TAT) = Completion Time (CT) - Arrival Time (AT)

Waiting Time (WT) = Turn Around Time (TAT) - Burst Time (BT)

Total TAT = 13 + 11 + 3 + 6 + 10 = 43

Total WT = 8 + 8 + 2 + 4 + 7 = 29

Average Waiting Time = 29 / 5 = 5.8 units

Average Turn Around Time = 43 / 5 = 8.6 units